Computer Methods in Biomechanics and Biomedical Engineering

ISSN: 1025-5842 (Print) 1476-8259 (Online) Journal homepage: www.tandfonline.com/journals/gcmb20

Reasons why dynamic compression plates are inferior to locking plates in osteoporotic bone: a finite element explanation

Alisdair R. MacLeod, A. Hamish R.W. Simpson & Pankaj Pankaj

To cite this article: Alisdair R. MacLeod, A. Hamish R.W. Simpson & Pankaj Pankaj (2015) Reasons why dynamic compression plates are inferior to locking plates in osteoporotic bone: a finite element explanation, Computer Methods in Biomechanics and Biomedical Engineering, 18:16, 1818-1825, DOI: 10.1080/10255842.2014.974580

To link to this article: https://doi.org/10.1080/10255842.2014.974580

Published online: 04 Dec 2014.

Computer Methods in Biomechanics and Biomedical Engineering, 2015
Vol. 18, No. 16, 1818–1825, http://dx.doi.org/10.1080/10255842.2014.974580

Reasons why dynamic compression plates are inferior to locking plates in osteoporotic bone: a finite element explanation

Alisdair R. MacLeod[a]*, A. Hamish R.W. Simpson[b1] and Pankaj Pankaj[a2]

[a]*School of Engineering, The University of Edinburgh, Edinburgh EH9 3JL, Scotland, UK;* [b]*Department of Orthopaedic Surgery, The University of Edinburgh, Edinburgh EH16 4SU, Scotland, UK*

(*Received 25 March 2014; accepted 6 October 2014*)

While locking plate fixation is becoming increasingly popular for complex and osteoporotic fractures, for many indications compression plating remains the standard choice. This study compares the mechanical behaviour of the more recent locking compression plate (LCP) device, with the traditional dynamic compression plates (DCPs) in bone of varying quality using finite element modelling. The bone properties considered include orthotropy, inhomogeneity, cortical thinning and periosteal apposition associated with osteoporosis. The effect of preloads induced by compression plating was included in the models. Two different fracture scenarios were modelled: one with complete reduction and one with a fracture gap. The results show that the preload arising in DCPs results in large principal strains in the bone all around the perimeter of the screw hole, whereas for LCPs large principal strains occur primarily on the side of the screw proximal to the load. The strains within the bone produced by the two screw types are similar in healthy bone with a reduced fracture gap; however, the DCP produces much larger strains in osteoporotic bone. In the presence of a fracture gap, the DCP results in a considerably larger region with high tensile strains and a slightly smaller region with high compressive strains. These findings provide a biomechanical basis for the reported improved performance of locking plates in poorer bone quality.

Keywords: screw; locking compression plate; dynamic compression plate; principal strains; preload

Introduction

The types of internal fixation plating most commonly used differ primarily in the screw design. The dynamic compression plate (DCP) uses compression screws that tighten the plate to the bone. A newer type of plate, the locking compression plate (LCP), uses locking screws that have a threaded head enabling the screw to fasten into the plate as well as into the bone (Perren 2002). This connection means that the plate can be offset from the bone, which can help preserve the periosteum (Perren 2002). Compression screws can be mixed with locking screws when using the LCP; however, this study considers the exclusive use of each screw type.

Studies have suggested that the LCP performs better than the DCP in older or weaker bone (Gardner et al. 2004; Kim et al. 2007; Yanez et al. 2010). Locked plating has been reported to have increased fatigue strength and ultimate failure loads (Seide et al. 2007; Uhl et al. 2008; Zehnder et al. 2009). On the other hand, studies have shown that the pull-out strength of compression screws increases with bone density (DeCoster et al. 1990; Thiele et al. 2007), and they perform better than locked plating in healthy bone (Kim et al. 2007).

The reasons for the differences in failure loads between the plating types are not straightforward as there are multiple factors to consider. The preloads involved in compression screw tightening increase stress levels at the screw–bone interface even before physiological loads are applied, whereas locking screws have negligible screw-tightening preload (Gardner et al. 2004). During physiological loading, however, the DCP allows for frictional load transfer at the plate–bone interface; locked plating, on the other hand, transfers all physiological loads via the screw–bone interface. Therefore, the local environment around screws in a DCP is influenced by a mix of effects, whereas the mechanics related to locking screws is more straightforward – similar to a unilateral external fixation device.

Many experimental tests have been conducted comparing locking and compression screws (Kim et al. 2007; Seide et al. 2007; Uhl et al. 2008; Yanez et al. 2010). Although the experimental tests have been able to compare the performances, they have not been able to fully explain the mechanics associated with the differences. This study uses finite element simulation, which permits evaluation of the local stress/strain response around screws; this is difficult if not impossible to measure in a laboratory experiment.

A few computational simulations have been attempted as DCPs are complex to model due to the screw insertion producing axial preload in the screw (pulling the plate towards the bone) and interfragmentary compression caused by preload within the plate (compression screws

*Corresponding author. Email: a.macleod@ed.ac.uk

are inserted off-centre in plate holes that forces the bone towards the fracture gap) (Gardner et al. 2004). While compression plate preloads have been included in some previous numerical studies (Rybicki and Simonen 1977; Cheal et al. 1985; Beaupre et al. 1988; Ferguson et al. 1996; Kim et al. 2010), their inclusion is via application of concentrated loads or as prescribed displacements. The effect produced by such techniques is not completely realistic; for example, an applied displacement to the bottom of a screw does not induce loads akin to screw fastening as the threads closest to the displacement will carry larger loads. Moreover, screw threads are rarely modelled and can have considerable influence on the local strain environment. Gefen (2002) noted the high proportion of load taken by the first thread in a study where screw-tightening preload was modelled.

It has been previously shown that the fracture gap size affects the strain within the plate (Stoffel et al. 2003; Oh et al. 2010); however, to our knowledge, the influence of a fracture gap on the strains in the bone around screw holes has not been investigated. Aseptic loosening of screws is a significant clinical problem that is generally attributed to over or under exposure to strain (Huiskes et al. 1985; Evans et al. 1990; Turner et al. 1997; Stoffel et al. 2003). Moreover, this threshold may be below the yield strain of bone (Sugiura et al. 2000) and osteoporotic bone is thought to be more susceptible to loosening than healthy bone (Gardner et al. 2004). High levels of strain around screw holes are therefore strong indicators of potential loosening.

This study models the mechanical environment in DCP fixation in a more realistic manner. The local strain environment around screw holes and the stresses within the implants are predicted for varying bone qualities and fracture patterns. The results are compared with those for a locking plate. The aim of the study was to provide a biomechanical basis for the reported improved performance of locking plates in poorer bone quality. The study also aimed to examine the modelling approaches used to represent screw fastening for DCPs.

Methods

For both compression plating and locked plating, models were developed to represent a fracture gap scenario where no interfragmentary contact can occur and for a fully reduced fracture. The DCP is not generally used as a bridge plate; however, complete reduction of the fracture may not always be achieved leaving a fracture gap. Both plating systems and fracture gap scenarios were modelled for healthy and osteoporotic bone using representative cross-sectional geometries and material properties. As the study examined the tibial diaphysis, only the cortical bone was included in the models.

Two reference tibial cross sections with cross-sectional areas of 319 and 265 mm^2 with average unicortical thicknesses of 5.1 and 3.64 mm (Russo et al. 2006; Donaldson et al. 2012b) were considered for healthy and osteoporotic bone. These tibial cross sections were extruded longitudinally to a length of 110 mm for three-dimensional finite element modelling (Figure 1(a)–(d)). We considered a transverse fracture of the tibial diaphysis and assumed symmetry on either side of the fracture (Figure 1(c)). Locking plates are generally used with an offset between the plate and the bone (Figure 1(b)). A previous study (Ahmad et al. 2007) investigated the influence of offset and found that 0–2 mm offsets performed similarly in terms of load–deformation response, while values greater than this were detrimental. Therefore, we assumed no interaction between these two components and a 2-mm offset in the locking plate models. Callus or soft tissue was not included in any of the models.

The plate dimensions were the same for the DCP and the LCP and based on typical industry standard: nine-hole plate with a cross section of 3.6 mm × 12 mm; hole spacing of 20 mm. The screws had an external diameter of 4.5 mm and core diameters of 3.84 and 3.5 mm for locking and compression screws, respectively. Threads, with a spacing of 1.4 mm, were modelled as idealised rings rather than helical, an approach that has been previously used (Donaldson et al. 2012a; MacLeod et al. 2012). This assumption eased meshing and was thought to be unlikely to affect the results. For simplification, only the threads in contact with the bone were modelled.

Screws were placed in the second and fourth holes from the fracture gap (Figure 1(d)). The models used approximately 220,000 linear tetrahedral elements with

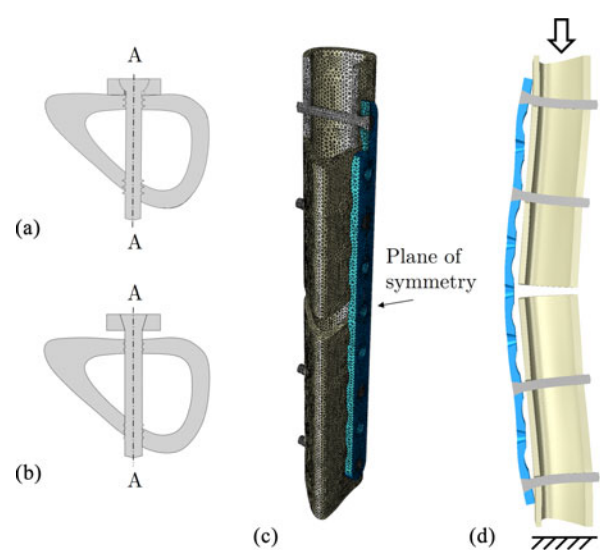

Figure 1. (a) Cross section of a healthy tibia with DCP, (b) cross section with LCP, (c) finite element model geometry of the LCP with section through the centre line of the plate showing the mesh refinement around screw holes and (d) deformed shape of the LCP model (scale factor × 40).

refinement around screw holes (Figure 1(c)). In the region around screw holes, the average element edge length was 0.3 mm. A mesh convergence study was performed and the results showed that doubling the mesh density of the plate increased the displacement at the fracture gap by 6.78% in healthy bone and 6.85% in osteoporotic bone. In the bone, the change in the peak minimum (and maximum) compressive strain was less than 3.22% (3.07%) when doubling the mesh resolution. Geometrically nonlinear analyses were conducted using Abaqus 6.10 (Simulia, Providence, RI, USA).

The cortical bone properties used were linear elastic, orthotropic and heterogeneous (variable through the cortical thickness) and are shown in Table 1. Two groups of bone properties used were representative of female cortical bone and based on a previous study (Donaldson et al. 2012b). The anisotropy was incorporated using cylindrical axes for each segment of the cross section (Figure 2(a)), where E11 is the radial, E22 is the circumferential and E33 is the longitudinal direction. Table 1 provides values at the periosteum and the endosteumn (Donaldson et al. 2012b); the variation in orthotropic elastic properties through the thickness was evaluated using tensor interpolations (Yang et al. 1999) and is illustrated in Figure 2(b). Cortical thinning and periosteal apposition were also incorporated for osteoporotic models as shown in Figure 2(c) (Russo et al. 2006; Donaldson et al. 2012b). The plate and screws were assumed to be homogeneous, isotropic and linear elastic steel. The Young's modulus for both was 1.80×10^{11} N/m^2 (Gefen 2002; Ganesh et al. 2005; Benli et al. 2008; Fouad 2010). Poisson's ratio for steel was taken as 0.3 (Ganesh et al. 2005; Benli et al. 2008; Oh et al. 2010).

Both linear and nonlinear models have previously been considered for friction between the bone and the implant (Shirazi-Adl et al. 1993; Dammak et al. 1997; Eser et al. 2010; Fouad 2010; Karunratanakul et al. 2010; Pessoa et al. 2010). The screw–bone interaction at the entrant cortex was modelled as frictional using a standard Coulomb friction coefficient of 0.3 based on some of the recent studies (Eser et al. 2010; Pessoa et al. 2010; MacLeod et al. 2012). As previously found with unilateral fixators (Donaldson et al. 2012a), the peak strains were found to be concentrated around the screw holes on the entrant cortex; to simplify simulation the screw–bone interaction at the exit cortex was modelled as tied. The plate–bone interaction for DCP used a frictional coefficient of 0.4 (Perren 2002; Kim et al. 2010).

The load was applied using uniform displacement control at the free end as shown in Figure 1(d). The analysis was conducted up to a load of 1200 N (approximately 1.6 times bodyweight for a 75-kg person), which has been shown to be the force in the tibial mid-shaft at 7 weeks post-surgery (Vijayakumar et al. 2006).

The screw-fastening preload was simulated using 'bolt load' in Abaqus 6.10. This preload was applied to a slice of the screw shaft, below the head and above the first thread (Figure 3(a)). This novel application induces normal stresses in the axial direction of the screw and does not require any artificial constraints or prescribed displacements. Values of preload vary widely in the literature; a value of 500 N was chosen based on an average of some

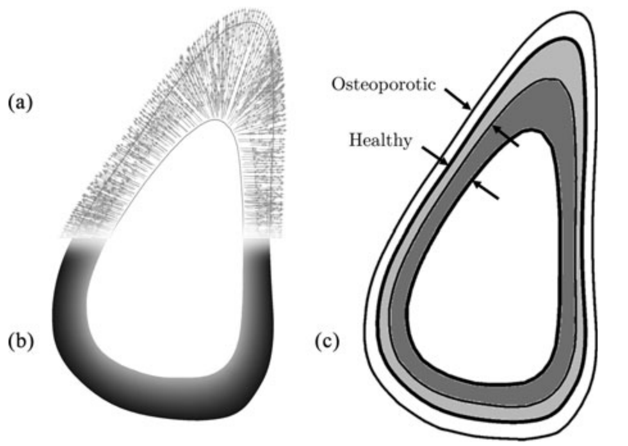

Figure 2. The material properties incorporated showing (a) orthotropic material orientations, (b) heterogeneity – gradient from endosteal to periosteal surfaces and (c) the geometrical changes – periosteal apposition and endocortical resorption associated with osteoporosis.

Table 1. Material properties of the bone.

	Young/healthy		Old/osteoporotic	
GPa	Periosteum	Endosteum	Periosteum	Endosteum
E11	18.6	16.6	12.9	3.2
E22	18.8	17.1	14.6	6.0
E33	22.4	21.4	19.3	11.2
G12	7.2	6.6	5.4	1.8
G13	6.9	6.4	5.4	2.2
G23	7.0	6.5	5.7	3.0
v12	0.29	0.27	0.24	0.16
v13	0.26	0.24	0.20	0.07
v23	0.26	0.24	0.22	0.14

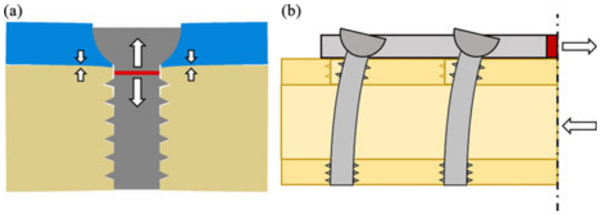

Figure 3. Depictions of the preloads used in the DCP models: (a) screw tightening and resulting bone–plate compression; (b) plate tensioning and resulting interfragmentary compression. The red regions are where the preload was specified and white arrows indicate the direction of the preloads.

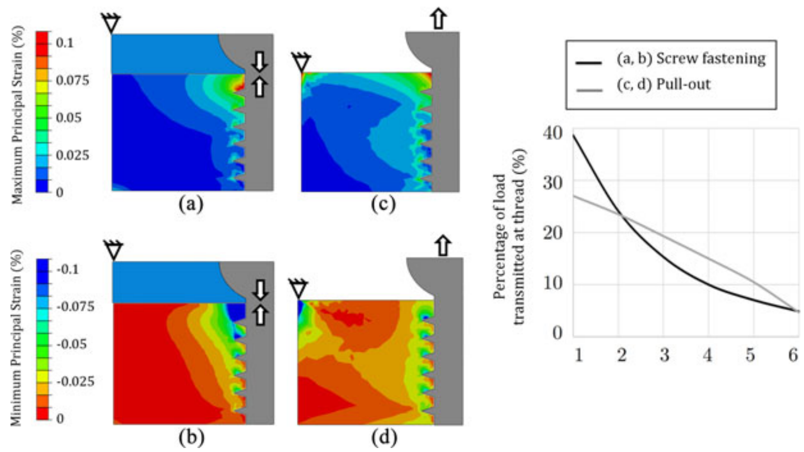

Figure 4. Maximum and minimum principal strains in idealised models showing: (a, b) the bolt preload representation of screw fastening with a plate; (c, d) pull-out style loading, and the percentage load transferred at each thread counting from the screw head for the two loading cases.

previous studies (Rybicki and Simonen 1977; Zand et al. 1983; Cheal et al. 1985; Beaupre et al. 1988; Perren 2002; Kim et al. 2010; Ya'ish et al. 2011). In the DCP models where the fracture was completely reduced, this approach was also used to simulate a 500 N tensile preload in the plate (Figure 3(b)). As radial interference preload is likely to be present in both screw types (MacLeod et al. 2012) it was not included to better highlight the differences between the two plating systems.

Results

Many studies use pull-out tests to evaluate the performance of bone screws (Feerick and McGarry 2012); however, concerns have been raised as to whether they are representative of screw fastening (Gefen 2002). An additional idealised simulation was run to demonstrate the differences in mechanical response between pull-out and screw-fastening, particularly in terms of stress–strain environment. The simulations used a pull-out force identical to the fastening force, and it was applied as a concentrated load. The loading and boundary conditions for both tests are shown in Figure 4. The boundary conditions used for pull-out are similar to previous studies with restraint around the perimeter of the body of interest. In both cases, the largest strains occur in the vicinity of the first screw thread, which also transmits the most load (Figure 4). It is clear that the strain distribution produced by pull-out is distinct from screw fastening with a plate.

For the full bone–plate construct models, minimum and maximum principal strain contours were plotted using section A–A (Figure 1(a),(b)) in the vicinity of the screw hole on the proximal side of the fracture (Figures 5 and 6).

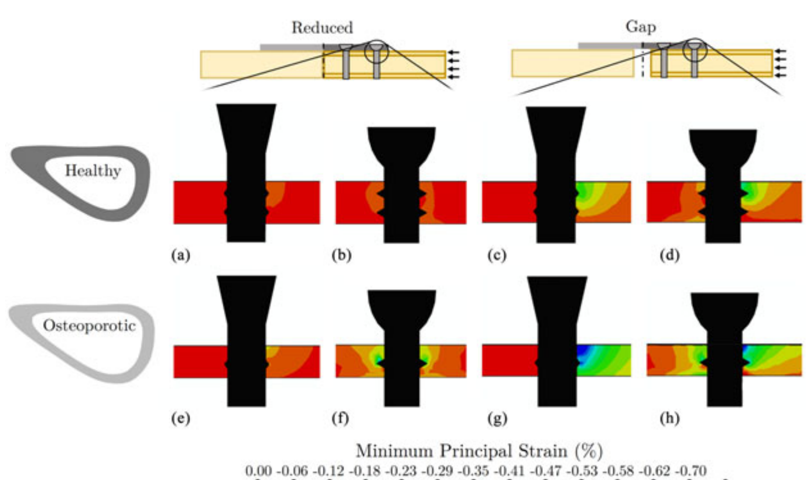

Figure 5. Minimum principal strain around the screw hole furthest from the fracture gap in the LCP (a, c, e, g) and DCP (b, d, f, h) taking a transverse section through the centreline of the plate (section A–A in Figure 1).

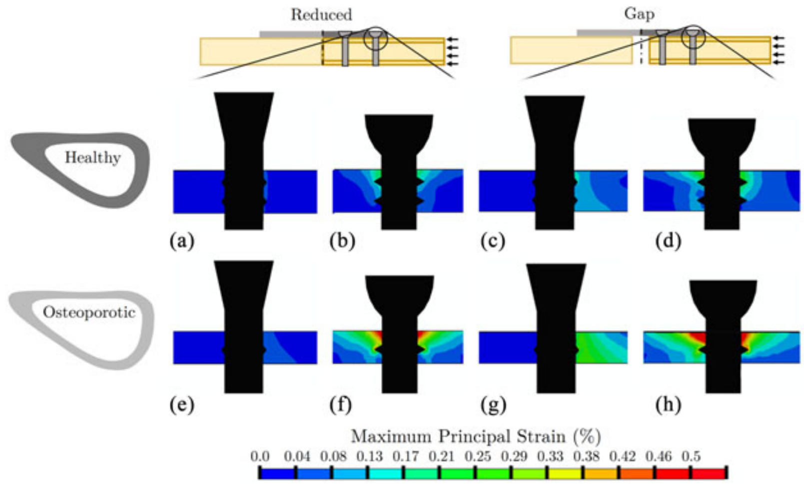

Figure 6. Maximum principal strain around the screw hole furthest from the fracture gap in the LCP (a, c, e, g) and DCP (b, d, f, h) taking a transverse section through the centreline of the plate (section A–A in Figure 1).

The plots compare strains in the LCP (a, c, e, g), the DCP (b, d, f, h), healthy bone (a, b, c, d), osteoporotic bone (e, f, g, h), a fully reduced fracture (a, b, e, f) and a fracture gap (c, d, g, h). Principal strain rather than stress was used for the comparisons (Pankaj 2013) and the limits of the scale were chosen to be 0.5% and 0.7% for tensile and compressive strains, respectively. These values represent yield values for cortical bone found in the literature (Bayraktar et al. 2004; Ebacher et al. 2007). For the DCP, large compressive and tensile strains occur around the full perimeter of the DCP screws (particularly apparent in Figures 5(f) and (h) and 6(f) and (h)). These arise due to the screw-tightening preload. Large principal strains in the LCP are primarily confined to the loaded side of the perimeter (Figures 5(c) and (g) and 6(c) and (g)). For the LCP, these arise due to screw–bone interface compression, in the direction of the load, and consequent circumferential tension and pull-out effects. For both types of plating, larger principal strains are located in the periosteal region; however, this effect is more pronounced for DCP, particularly at the periosteal side of the first screw thread (Figures 5(h) and 6(h)).

The volume of the bone with maximum (and minimum) principal strain greater than 0.2% (−0.2%), termed MaxEV (and MinEV), was determined for each of the healthy (Figure 7(a)) and the osteoporotic (Figure 7(b)) bone models. This value is close to the upper limit of physiological strain, and therefore represents regions susceptible to resorption or loosening (Turner et al. 1997; Sugiura et al. 2000). In healthy bone with a reduced fracture, both plating types result in negligible MinEV and MaxEV (Figure 7(a)); this is also apparent from Figures 5 (a),(b) and 6(a),(b). However, for osteoporotic bone with reduced fracture gap, MaxEV and MinEV are substantially

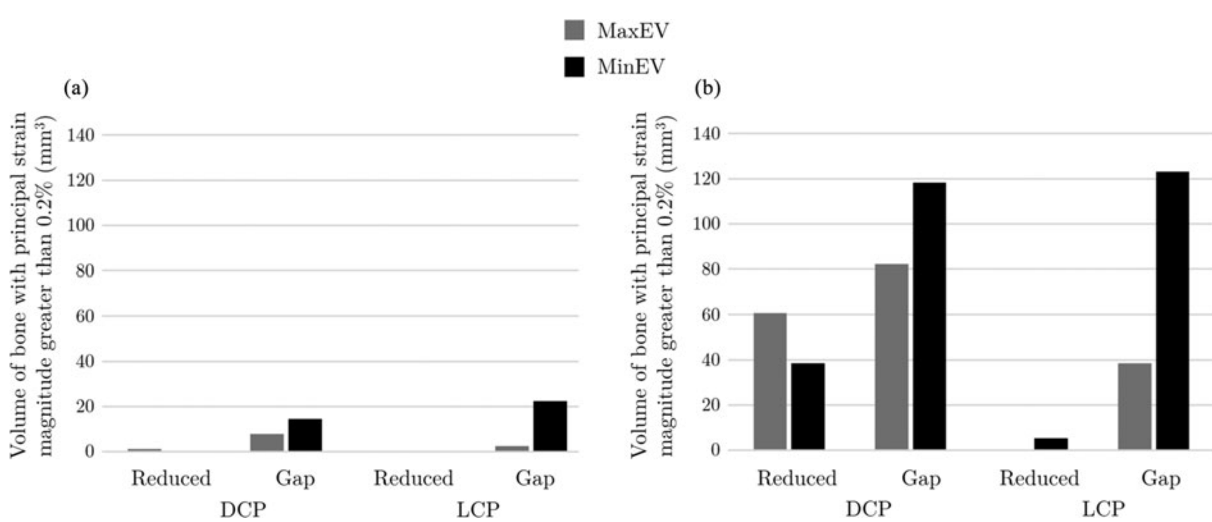

Figure 7. Volume of bone with principal strain magnitude greater than 0.2% for (a) healthy bone and (b) osteoporotic bone.

larger for DCP in comparison with LCP (Figure 7(b)); an effect also apparent from Figures 5(e),(f) and 6(e),(f). Both types of plating produced larger regions of high strain in the presence of a fracture gap (Figure 7), wherein DCP results in larger tensile region (MaxEV) and a slightly smaller compressive region (MinEV). For healthy bone with a fracture gap, the LCP produced 35% more MinEV and 67% less MaxEV than the DCP (Figure 7(a)). For osteoporotic bone with a fracture gap, the LCP produced 4% more MinEV than the DCP (Figure 7(b)), while the MaxEV region was found to be 53% smaller and restricted to the side of the screw hole proximal to the load (Figure 6 (g),(h)). The DCP produced larger regions of MaxEV than the LCP in all cases and this was most pronounced in osteoporotic bone (Figure 7(b)).

Geometrical and material properties were altered between the two bone qualities; the influence of these changes was examined independently to assess their relative importance. For the DCP, incorporating the geometrical changes of osteoporosis (but maintaining healthy material properties) produced MaxEV and MinEV values of 50% and 75% of the combined effect; with material changes alone, these values were 71% and 39%. Both effects, therefore, are important for correct prediction of strain.

Discussion

The results show that the strain environment at the screw–bone interface explains the superior performance of locked plating in osteoporotic bone. Compared with osteoporotic bone, healthy bone produces much smaller regions of high strain in both plate types.

The bolt preload representation of screw fastening used in this study was found to produce results similar to a study by Gefen (2002) and to the analytical models of Grewal and Sabbaghian (1997). Screw-tightening results in compression of the bone against the plate; this increases the proportion of load taken in the vicinity of the first thread compared with pull-out style loading.

This study shows that locking plates produce lower tensile strains around screw-hole locations in osteoporotic bone than DCPs. As locked plating produced similar compressive strains in gap situations, it can be inferred that it is the large tensile strains produced by the DCP that are problematic.

The DCP has angled screw holes in the plate to induce a preload between fracture fragments during screw insertion (Gardner et al. 2004) and are implanted using open surgery with direct bone exposure. This aids in fracture reduction and helps eliminate any gaps between fractured bone ends. Reduction can be harder to achieve using the LCP, particularly when minimally invasive techniques are being used. In both types of plating, an incomplete fracture reduction or a fracture gap resulted in increased strain concentration around screw holes. These results support clinical conclusions that simple fractures should be treated with reduction and absolute stability (Leahy 2010). Compression-plated osteoporotic bone, however, resulted in regions of very high tensile strain, even with complete fracture reduction.

While strains due to weight bearing may be reduced by the addition of more screws (Donaldson et al. 2012a), the strains caused by screw fastening in DCPs will be present regardless of the number of screws used. Screw fastening alone accounted for 74% of the MaxEV in osteoporotic bone and 36% in healthy bone. The MinEV produced was around 45% for both bone qualities. Another major difference in strain distribution between the screw types was that locked plating mainly produced strains on the side of the screw proximal to the load; compression plating produced strains all around the perimeter. Screw-tightening preloads are therefore fundamental to modelling the strain environment produced by the DCP.

While screw fastening induced large strains in osteoporotic bone, the volume of the bone exceeding the yield strain due to screw fastening alone was less than $2\,mm^3$. Therefore, failure of the screw–bone interface would not be expected during operative screw insertion; however, the addition of physiological loads increased this volume to approximately $10\,mm^3$.

In the reduced fracture pattern, interfragmentary preload was included in the DCP models. This increased the proportion of load transmitted through the bone via interfragmentary contact compared with locking plates. In reduced fractures, therefore, an LCP shields the bone from load at the interfragmentary contact location by redirecting it through the plate. This aspect of compression plating, however, only applies when the fracture is fully reduced. In models with a fracture gap, the strains were largest in the periosteal surface in the screw entrant cortex (Figure 5(g),(h)). This is similar to strain distributions seen in external fixators (Huiskes et al. 1985; Donaldson et al. 2012a).

This study has some limitations. The preloads induced by the DCP will dissipate over time due to remodelling, damage or the viscoelasticity of bone. In addition, the rate of the dissipation may vary with bone quality. Our models, however, were not equipped to predict such behaviour. In an ovine study, Hyldahl et al. (1991) found that different preload types influenced the amount of resorbed bone at 5 weeks. This demonstrates that the effects of initially applied preload are measureable well into the healing period.

It is also likely that surgeons would not be able to fasten the screws as tightly in osteoporotic bone and therefore the screw tensile preload would be smaller than in healthy bone. To make the models comparable, however, the same value was used in this study.

Material nonlinearity was not included in the models. This has been shown to have a negligible influence on fracture gap motion (Donaldson et al. 2012a) but could influence the stress–strain environment around screw holes. Nevertheless, even in the model with the largest strains, only 0.01% of the volume of the bone was above the compressive yield strain (0.7%) and 0.03% above the tensile yield strain (0.5%). Even with material nonlinearity included, the locations of peak strain are likely to remain unchanged.

Finite element modelling can provide information such as strains around screw holes that can be difficult or impossible to measure in a physical experiment. This study predicted that locked plating produced far smaller tensile strains around screw holes in osteoporotic bone than the DCP. This provides a mechanical basis for the improved performance of locking plates in poorer bone quality and explains previously reported higher incidence of screw loosening using the DCP (Bottlang et al. 2009). These differences in strain at the screw–bone interface were achieved using more realistic modelling of screw fastening in the DCP. This will be important in any further studies with an aim to understand bone remodelling, fatigue behaviour or loosening of screws.

Conflict of interest disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The authors gratefully acknowledge the support of Orthopaedic Research UK [grant number RP 10-48].

Notes

1. Email: hamish.simpson@ed.ac.uk
2. Email: pankaj@ed.ac.uk

References

Ahmad M, Nanda R, Bajwa AS, Candal-Couto J, Green S, Hui AC. 2007. Biomechanical testing of the locking compression plate: when does the distance between bone and implant significantly reduce construct stability? Injury. 38(3):358–364.

Bayraktar HH, Morgan EF, Niebur GL, Morris GE, Wong EK, Keaveny TM. 2004. Comparison of the elastic and yield properties of human femoral trabecular and cortical bone tissue. J Biomech. 37(1):27–35.

Beaupre GS, Carter DR, Orr TE, Csongradi J. 1988. Stresses in plated long-bones: the role of screw tightness and interface slipping. J Orthop Res. 6(1):39–50.

Benli S, Aksoy S, Havitcioglu H, Kucuk M. 2008. Evaluation of bone plate with low-stiffness material in terms of stress distribution. J Biomech. 41(15):3229–3235.

Bottlang M, Doornink J, Byrd GD, Fitzpatrick DC, Madey SM. 2009. A nonlocking end screw can decrease fracture risk caused by locked plating in the osteoporotic diaphysis. J Bone Joint Surg Am. 91:620–627.

Cheal EJ, Hayes WC, White AA, Perren SM. 1985. Stress-analysis of compression plate fixation and it's effects on long-bone remodeling. J Biomech. 18(2):141–150.

Dammak M, ShiraziAdl A, Zukor DJ. 1997. Analysis of cementless implants using interface nonlinear friction – experimental and finite element studies. J Biomech. 30(2):121–129.

DeCoster TA, Heetderks DB, Downey DJ, Ferries JS, Jones W. 1990. Optimizing bone screw pullout force. J Orthop Trauma. 4(2):169–174.

Donaldson FE, Pankaj P, Simpson AHRW. 2012a. Bone properties affect loosening of half-pin external fixators at the pin-bone interface. Injury. 43(10):1764–1770.

Donaldson FE, Pankaj P, Simpson AHRW. 2012b. Investigation of factors affecting loosening of ilizarov ring–wire external fixator systems at the bone–wire interface. J Orthop Res. 30(5):726–732.

Ebacher V, Tang C, McKay H, Oxland TR, Guy P, Wang RZ. 2007. Strain redistribution and cracking behavior of human bone during bending. Bone. 40(5):1265–1275.

Eser A, Tonuk E, Akca K, Cehreli MC. 2010. Predicting time-dependent remodeling of bone around immediately loaded dental implants with different designs. Med Eng Phys. 32(1):22–31.

Evans M, Spencer M, Wang Q, White SH, Cunningham JL. 1990. Design and testing of external fixator bone screws. J Biomed Eng. 12(6):457–462.

Feerick EM, McGarry JP. 2012. Cortical bone failure mechanisms during screw pullout. J Biomech. 45(9):1666–1672.

Ferguson SJ, Wyss UP, Pichora DR. 1996. Finite element stress analysis of a hybrid fracture fixation plate. Med Eng Phys. 18(3):241–250.

Fouad H. 2010. Effects of the bone-plate material and the presence of a gap between the fractured bone and plate on the predicted stresses at the fractured bone. Med Eng Phys. 32(7):783–789.

Ganesh VK, Ramakrishna K, Ghista DN. 2005. Biomechanics of bone-fracture fixation by stiffness-graded plates in comparison with stainless-steel plates. Biomed Eng Online. 4(46).

Gardner MJ, Helfet DL, Lorich DG. 2004. Has locked plating completely replaced conventional plating? Am J Orthop (Belle Mead, NJ). 33(9):439–446.

Gefen A. 2002. Optimizing the biomechanical compatibility of orthopedic screws for bone fracture fixation. Med Eng Phys. 24(5):337–347.

Grewal AS, Sabbaghian M. 1997. Load distribution between threads in threaded connections. J Pres Vessel Technol. 119(1):91–95.

Huiskes R, Chao EYS, Crippen TE. 1985. Parametric analyses of pin-bone stresses in external fracture fixation devices. J Orthop Res. 3(3):341–349.

Hyldahl C, Pearson S, Tepic S, Perren SM. 1991. Induction and prevention of pin loosening in external fixation: an *in vivo* study on sheep tibiae. J Orthop Trauma. 5(4):485–492.

Karunratanakul K, Schrooten J, Van Oosterwyck H. 2010. Finite element modelling of a unilateral fixator for bone reconstruction: importance of contact settings. Med Eng Phys. 32(5):461–467.

Kim SH, Chang SH, Jung HJ. 2010. The finite element analysis of a fractured tibia applied by composite bone plates considering contact conditions and time-varying properties of curing tissues. Compos Struct. 92(9):2109–2118.

Kim T, Ayturk UM, Haskell A, Miclau T, Puttlitz CM. 2007. Fixation of osteoporotic distal fibula fractures:

a biomechanical comparison of locking versus conventional plates. J Foot Ankle Surg. 46(1):2–6.

Leahy M. 2010. When locking plates fail. AAOS Now. May 2010 Issue. Available from: www.aaos.org/news/aaosnow/may10/clinical4.asp

MacLeod AR, Pankaj P, Simpson AHRW. 2012. Does screw-bone interface modelling matter in finite element analyses? J Biomech. 45(9):1712–1716.

Oh JK, Sahu D, Ahn YH, Lee SJ, Tsutsumi S, Hwang JH, Jung DY, Perren SM, Oh CW. 2010. Effect of fracture gap on stability of compression plate fixation: a finite element study. J Orthop Res. 28(4):462–467.

Pankaj P. 2013. Patient-specific modelling of bone and bone-implant systems: the challenges. Int J Numer Methods Biomed Eng. 29(2):233–249.

Perren SM. 2002. Evolution of the internal fixation of long bone fractures. J Bone Joint Surg Br. 84B(8):1093–1110.

Pessoa RS, Muraru L, Marcantonio E, Vaz LG, Sloten JV, Duyck J, Jaecques SVN. 2010. Influence of implant connection type on the biomechanical environment of immediately placed implants – CT-based nonlinear, three-dimensional finite element analysis. Clin Implant Dent Relat Res. 12(3): 219–234.

Russo CR, Lauretani F, Seeman E, Bartali B, Bandinelli S, Di Iorio A, Guralnik J, Ferrucci L. 2006. Structural adaptations to bone loss in aging men and women. Bone. 38(1):112–118.

Rybicki EF, Simonen FA. 1977. Mechanics of oblique fracture fixation using a finite-element model. J Biomech. 10(2): 141–148.

Seide K, Triebe J, Faschingbauer M, Schulz AP, Puschel K, Mehrtens G, Jurgens C. 2007. Locked vs. unlocked plate osteosynthesis of the proximal humerus – a biomechanical study. Clin Biomech. 22(2):176–182.

Shirazi-Adl A, Dammak M, Paiement G. 1993. Experimental determination of friction characteristics at the trabecular bone/porous-coated metal interface in cementless implants. J Biomed Mater Res. 27(2):167–175.

Stoffel K, Dieter U, Stachowiak G, Gächter A, Kuster MS. 2003. Biomechanical testing of the LCP – how can stability in locked internal fixators be controlled? Injury. 34(Supplement 2):11–19.

Sugiura T, Horiuchi K, Sugimura M, Tsutsumi S. 2000. Evaluation of threshold stress for bone resorption around screws based on *in vivo* strain measurement of miniplate. J Musculoskelet Neuronal Interact. 1(2):165–170.

Thiele OC, Eckhardt C, Linke B, Schneider E, Lill CA. 2007. Factors affecting the stability of screws in human cortical osteoporotic bone: a cadaver study. J Bone Joint Surg Br. 89(5):701–705.

Turner CH, Anne V, Pidaparti RM. 1997. A uniform strain criterion for trabecular bone adaptation: do continuum-level strain gradients drive adaptation? J Biomech. 30(6): 555–563.

Uhl JM, Seguin B, Kapatkin AS, Schulz KS, Garcia TC, Stover SM. 2008. Mechanical comparison of 3.5 mm broad dynamic compression plate, broad limited-contact dynamic compression plate, and narrow locking compression plate systems using interfragmentary gap models. Vet Surg. 37(7):663–673.

Vijayakumar V, Marks L, Bremmer-Smith A, Hardy J, Gardner T. 2006. Load transmission through a healing tibial fracture. Clin Biomech. 21(1):49–53.

Ya'ish FMM, Nanu AM, Cross AT. 2011. Can DCP and LCP plates generate more compression? The effect of multiple eccentrically placed screws and their drill positioning guides. Injury. 42(10):1095–1100.

Yanez A, Carta JA, Garces G. 2010. Biomechanical evaluation of a new system to improve screw fixation in osteoporotic bones. Med Eng Phys. 32(5):532–541.

Yang GY, Kabel J, Van Rietbergen B, Odgaard A, Huiskes R, Cowin SC. 1999. The anisotropic Hooke's law for cancellous bone and wood. J Elast. 53(2):125–146.

Zand MS, Goldstein SA, Matthews LS. 1983. Fatigue failure of cortical bone screws. J Biomech. 16(5):305–311.

Zehnder S, Bledsoe JG, Puryear A. 2009. The effects of screw orientation in severely osteoporotic bone: a comparison with locked plating. Clin Biomech. 24(7):589–594.